

MultiMediate 2023: Engagement Level Detection using Audio and Video Features

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ABSTRACT

Real-time engagement estimation holds significant potential across various research areas, particularly in the realm of human-computer interaction. It empowers artificial agents to dynamically adjust their responses based on user engagement levels, fostering more intuitive and immersive interactions. Despite the strides in automating real-time engagement estimation, the task remains challenging in real-world settings, especially when handling multi-modal human social signals. Capitalizing on human body and audio signals, this paper explores the appropriate feature representations of different modalities and effective modelling of dual conversations. This results in a novel and efficient multi-modal engagement detection model. We thoroughly evaluated our method in the *MultiMediate'23* grand challenge. It performs consistently, with a notable improvement over the baseline model. Specifically, while the baseline achieves a concordance correlation coefficient (CCC) of 0.59, our approach yields a CCC of 0.70, suggesting its promising efficacy in real-life engagement detection.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

engagement, machine learning, neural networks

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1 INTRODUCTION

In dialogue, participants interact with each other, exchanging opinions and sharing stories. The level of participation and involvement of each participant in the interaction is referred to as "engagement." It is also defined as "the value that a participant in an interaction attributes to the goal of being together with the other participant(s) and of continuing the interaction" [18]. Estimating the value of engagement plays a vital role across various domains including education [26], healthcare [8] and UX optimization [3].

Traditional methods for evaluating engagement have predominantly relied on questionnaires and human observations, providing subjective measures of engagement [1, 25]. However, with the rapid advancement of machine learning (ML) techniques in recent years, a shift towards more objective and automated measures of engagement has been observed. The application of ML techniques to analyze engagement levels has emerged as a promising approach, offering the potential for real-time and accurate measurements of engagement [5, 12, 23].

Although the potential of utilizing ML and deep learning techniques to conduct engagement evaluation is recognized by researchers, it also presents several key challenges. Most of the existing methods are limited by their reliance on intrusive and external sensors. Their applicability thus is restricted in real-life scenarios [6, 12]. Among the studies that worked on non-intrusive signals-based engagement detection, many are constrained by depending solely on a single modality [7, 9, 13]. The few available multi-modal approaches [19, 22, 24], however, employed multi-modality without the fully understanding of the effective modelling and appropriate feature representations for different modalities. This hinders the

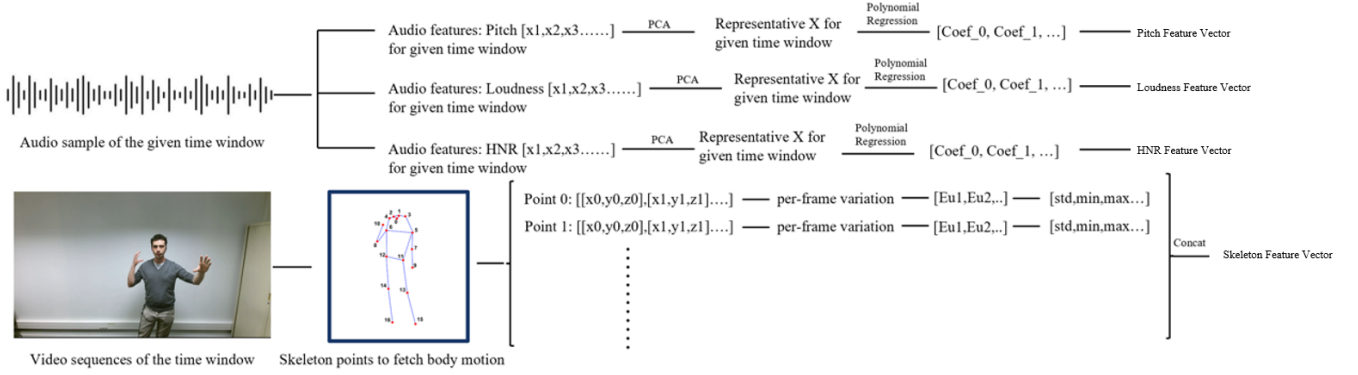


Figure 1: The overview of audio and video feature extraction for one participant.

development of more effective multi-modal approaches, especially for end-to-end and deep learning methods. Furthermore, it is believed that engagement is reflected not only on the target individual but also in the overall state of both conversation participants [21]. Yet, existing methods often only utilize information from the target individual, while overlooking his/her conversational partner(s).

To address these challenges, this paper examines the suitable feature representations of different modalities, particularly the human body motion and audio signals, and the adequate way of dual conversation modelling. We apply the optimal feature representations and modelling framework with feed-forward neural network (FFN) to yield an efficacious multi-modal engagement detection model.

We evaluate our method and conduct exhaustive ablation study in the *MultiMediate'23* engagement detection challenge. The experimental result indicate that *shape-based features* are more appropriate for audio signals, while *statistic-based features* are better suited for body motion signals in engagement detection. It also demonstrates the significance of modelling conversation partner's behavior. Finally, our model incorporated with the optimal features and modelling framework achieves a concordance coefficients correlation of 0.7, beating the baseline model, on the challenge test set [16]. This study not only contributes in engagement detection, but also paves the way for future studies in multi-modal and multi-partner conversation analysis.

2 DATASET AND EXPLORATORY ANALYSIS

MultiMediate'23 challenge [16] prepares a dataset for engagement detection based on the Novice-Expert Interaction (Noxi) database [4]. We conduct exploratory data analysis on the training set to design our method, and evaluate it on the validation and test set.

2.1 Engagement Detection and Noxi Dataset

Noxi is a novel database consisting of natural dual conversations between novice and expert participants [4]. These conversations were conducted in a video conference setting, in which participants focused on exchanging and retrieving information. The dataset has approximately 4TB data, including source video files and prefabricated multi-modality audio and video features. It also provides data that were captured using Microsoft Kinect2 [15]. Engagement level was annotated for each frame, which serves as the ground-truth labels for the MultiMediate'23 engagement detection challenge [16].

2.2 Exploratory Data Analysis

We conduct exploratory data analysis on the session recordings in the training set, to identify potential significant features for engagement detection. Specifically, we grab and output the lowest and highest engaging moments with a 5-second time window from each session. This yields a set of low and high engagement level clips respectively.

By observing the data, we find that participant engagement peaks when they are trying hard to convey specific information to the listener. This often occurs in situations where an expert is attempting to explain something to a novice, or when a novice expresses their understanding and poses questions. In many cases, it is accompanied with extensive body motion and a tone rich in emotion. On the other hand, low engagement occurs when the participant or her conversation partner become distracted by irrelevant tasks, such as an unexpected phone call or malfunctioning microphone. Subdued body movement and silence is observed in such situations.

3 METHODOLOGY

Based on the observations, we propose our features and model for engagement detection. In particular, we aim at extracting features that encode the emotion characteristics and the body motion summary from audio and video signals respectively. We further incorporate these features with *Target-Partner* dual conversation modelling to achieve engagement detection. Both the audio and video features are extracted from a given time window end at the detection moment (target frame).

3.1 Audio Features

Extensive research has demonstrated the significance of pitch and loudness signals in the perception of speakers' emotions [14]. On the other hand, Harmonics-to-Noise Ratio (HNR) signal holds importance for speech clarity, contributing to the listener's attention and engagement during conversation [20]. Hence, we particularly extract audio features from these signals for engagement detection.

For each signal, we apply Principle Component Analysis (PCA) on the raw signal to achieve dimensional reduction for more efficient feature encoding and modelling. Specifically, given a signal (e.g., a pitch signal) with a time window of T and feature dimension of n : $X = [(x_1^1, \dots, x_1^n), (x_2^1, \dots, x_2^n), \dots, (x_T^1, \dots, x_T^n)]$, we employ PCA on the features at each time step to achieve dimension reduced signal $\hat{X} = [\hat{x}_1, \hat{x}_2, \dots, \hat{x}_T]$. We aim to devise features that encode the

emotional characteristics in audio signals. It is proved that “*shape-based features*” which measures the overall structure pattern of the audio signals are effective in emotion encoding [17]. Hence, we apply polynomial regression to the audio signals after PCA-based pre-processing, to model and capture their shape pattern. A higher degree polynomial is chosen for the regression, given the complex shape patterns of our audio signals. The coefficients from these regressions then form the final feature vector, creating a comprehensive, multidimensional representation of the audio signal within the given time window. As shown in Fig. 1, we extract a feature vector for each of the three audio signals.

3.2 Video Features

Noxi database provides the data collected by Microsoft’s Kinect2 [4]. This include the participants’ skeleton points (i.e., body key points), from which, body motion patterns can be encoded. Specifically, Kinect2 captures up to 25 unique body key points, such as the head, neck, shoulders, elbows, hands, knees, and feet, offering a comprehensive view of a person’s posture, movement, and interactions with their surroundings. We extract video features based on the 3D positions of these points.

Based the exploratory observations, we are particularly interested in features that summarize participants’ body motions. To this end, we focus on extracting “*statistic-based features*” of body moving signals, as suggested in previous body motion analysis studies [10]. We calculate the frame-by-frame variations using Euclidean distances for each body key point. This enables us to quantify the their movement amount between frames, forming the body moving signals. After normalizing these variation signals (session-based min-max normalization), we extract six *statistic-based features* for each as shown in Table 1). The features provide a comprehensive summary of the per-frame variations of different body parts, capturing their movement amount, range and distribution across time, thus is well-suited for body motion modeling. We group the features extracted from all the body key points into one feature vector for further modelling, as illustrated in Fig. 1.

Table 1: The proposed statistic-based features for body motion

Feature	Description
min	Minimum value of variation
max	Maximum value of variation
std	Standard error of variation values
mean	Average value of variation
qt15	Value at the 15% quantile of variation range
qt75	Value at the 75% quantile of variation range

3.3 Model Design and Implementation

Our features are extensively engineered based on the exploratory study. We thus have substantial flexibility to apply simple feed-forward neural network (FNN) with fully-connected (FC) layers to achieve modelling. Fig. 2 depicts the structure of the proposed model. It first transforms and concatenates the three audio feature vectors into one embedding vector that has the same dimension with the video feature vector. This ensure the equal weights of the two modalities. The embedded audio feature vector and video feature vector are then further concatenated and fed into a FC layer to yield the multi-modal feature vector, indicating the joint audio and video (body) representations of one participant.

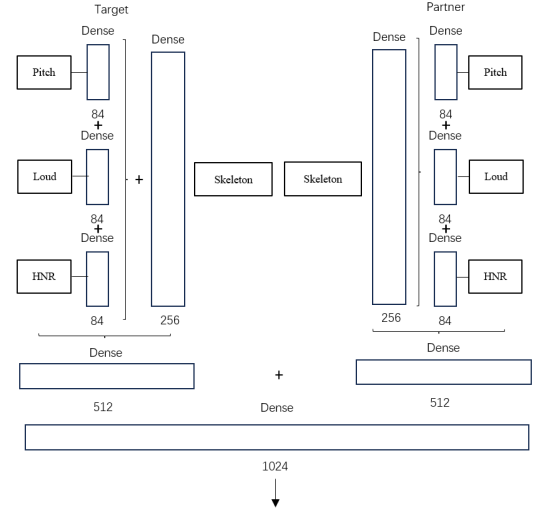


Figure 2: The proposed prediction model

Literature shows the significance of involving partners’ modelling in group conversation analysis [11]. In dual conversations, we also hypothesize that engagement is reflected not only on the target participant but also in the conversation partner and the overall interaction. Therefore, we design our method to jointly model the features from both the “*Target*” participant and his/her “*Partner*”. Finally, the *Target-Partner* concatenated feature vector is fed into the prediction layer to generate predictions. We utilize Adam as the optimizer with a learning rate of $1e-6$, and employ the mean-average error (MAE) as the loss function for model training.

4 EXPERIMENTAL RESULTS

We evaluate the proposed method and conduct comprehensively ablation study in the MultiMediate’23 engagement detection challenge [16]. We then apply our method with optimal feature and modelling framework on the challenge test set and compare with the competitors. We follow the challenge instruction to conduct frame-wise detection and use Concordance Correlation Coefficient (CCC) as evaluation metric.

4.1 Ablation Study

We examine the appropriate feature engineering, the contribution of multi-modal, the impact of time window size, and the proper dual conversation modelling in this paper.

4.1.1 Appropriate feature engineering. In the proposed model, we extract the *shape-based* features for audio signal encoding and *statistic-based* features for video (i.e., body motion) signal encoding. Moreover, we compute the frame-by-frame variations to highlight the movement nature of the body signals in our model. In the ablation study, we first evaluate the appropriate feature engineering methods for audio and video modalities respectively. This includes the examination of using *shape-based* or *statistic-based* features, and with or without the variation computation.

Table 2 and Table 3 shows the results (evaluated on the challenge validation set) for audio and video signals respectively. According to the results, *shape-based* features is better suited for audio signal encoding in the engagement detection task. *Shape-based* features

measure the overall shape and structure patterns of the audio signals, which effectively reveal their emotional attributes. They thus contribute more in engagement detection. This is consistent with the findings from prior studies [2, 17]. Moreover, when extract the *shape-based* features for audio signals, it is better to work on the raw inputs without considering the variation between frames. Applying audio channel alone with the optimal feature engineering achieves a promising CCC of 0.623 for engagement detection.

Table 2: Ablation study on audio features

Features	Variation Computing	Val CCC
<i>Shape-based</i>	Without	0.623
<i>Shape-based</i>	With	0.358
<i>Statistic-based</i>	Without	0.541
<i>Statistic-based</i>	With	0.423

For video channel, the results demonstrate the effectiveness of *statistic-based* features extracted from the body motion signals (i.e., the variations between consecutive frames for each body key points). *Statistic-based* is able to summarize the overall information of the body movements, including the movement amount, range, and distribution across the time, hence are more appropriate in engagement detection. This is also consistent with the conclusions in previous motion analysis studies [10].

Table 3: Ablation study on video features

Features	Variation Computing	Val CCC
<i>Shape-based</i>	Without	0.184
<i>Shape-based</i>	With	0.413
<i>Statistic-based</i>	Without	0.223
<i>Statistic-based</i>	With	0.455

4.1.2 Contribution of multi-modal. We then examine the the performance of multi-modal (audio and video) modelling, and compare with the models use solely one modality. The results illustrated in Table 4 indicate that using both modalities provide better performance (CCC of 0.732) than the models that use only one. Combining audio and video features offers more comprehensive information, capturing the holistic view of both speech characteristics and body movements of the participants. This analysis underscores the importance of utilizing multiple modalities in engagement estimation tasks. Furthermore, the audio-only model yields a better performance than the model utilizes video channel only, suggesting the significance of audio information in engagement detection.

4.1.3 Size of Time Window. Both the audio and video features are extracted from a given time window end at the detection moment (target frame). We further evaluate the effect of time window size applied for the feature extractions. Increasing the time window size from 5 seconds to 10 seconds resulted in improved system performance. This enhancement was evident in both the Validation CCC and Test CCC metrics as shown in Table 5. Utilizing longer time windows allowed for a more comprehensive representation of the data, capturing relevant information and underlying patterns over extended periods. The broader temporal range also helped mitigate the impact of short-term fluctuations and noise, resulting in a more stable feature set. Besides, longer window facilitates the identification of long-term dependencies and trends within the data.

Table 4: The contribution of multi-modal

Used Feature	Val CCC
Audio only	0.623
Video only	0.548
Audio + Video	0.732

Table 5: The impact of time window size

Used Feature	Time Window Size	Val CCC	Test CCC
Audio + Video	5s	0.732	0.664
Audio + Video	10s	0.741	0.68

4.1.4 Target-Partner Dual Conversation Modelling. It is believed that modelling the target participant together with his/her conversation partner can contribute to better performance in conversation analysis. We thus conduct study to determine the optimal setting for dual conversation modelling. We adjust the size of the FC layers for *Target* and *Partner* encoding to achieve *Target* and *Partner* feature vectors with different dimensions (256 or 512), and evaluate the performance of different combinations as shown in in Table 6. It worth noting that the model performs better when relying more on the partner’s features (i.e., partner feature vector has higher dimension). This suggests the significance of dual conversation modelling. Engagement is reflected not only on the target participant, it is even more important to model of the reactions and behavior of the conversation partner.

Table 6: Evaluation of *Target-Partner* dual conversation modelling

Target FD	Opponent FD	Val CCC	Test CCC
512	512	0.741	0.68
512	216	0.735	0.672
216	512	0.745	0.695

4.2 Testing performance

Finally, we apply our method with the optimal setting to the test set. It achieves a promising performance of about 0.7 CCC, outperforming the baseline model (0.59 CCC) presented in [16].

5 CONCLUSION

This paper presents an in-depth exploration on multi-modal engagement detection in dual conversations. We particularly examine the appropriate feature representations and engineering methods for audio and video (body) signals respectively, and the effective modelling framework for *Target-Partner* dual conversation modelling. This leads to an efficacious engagement detection model. With the optimal feature representations and modelling settings, our model yields a promising performance in the MultiMediate’23 engagement estimation challenge (0.7 CCC), beating the baseline model (0.59 CCC). Moving forward, our research will focus on developing model with more advanced neural networks like Transformer [27], with the aim of fully leveraging the valuable multi-modal and dual conversation information presented in this study.

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